Advancing Food Nutrition Estimation via Visual-Ingredient Feature Fusion

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ABSTRACT

Accurate estimation of food nutritional content from images is a key challenge in dietary monitoring and public health, yet progress has been hindered by the scarcity of large-scale, nutrition-annotated benchmarks and the failure of vision-only models to exploit the strong nutritional signal carried by ingredients. We introduce **FastFood**, a large-scale benchmark comprising 84,446 images across 908 fast-food categories from 10 global brands, annotated with calories, fat, carbohydrates, protein, and ingredient lists. To leverage ingredient information at scale, we propose **VIF**^{2} (Visual-Ingredient Feature Fusion), a model-agnostic framework that encodes ingredient lists using a frozen CLIP text encoder, projects the resulting embeddings into the visual feature space, and fuses them into intermediate CNN feature maps via channel-wise addition or into ViT representations as an additional token. At test time, where ground-truth ingredients are unavailable, we employ a large multimodal model (LMM) to predict ingredients from multiple augmented views with majority voting. Experiments on FastFood and the Nutrition5k benchmark demonstrate substantial improvements: VIF^2 reduces the caloric MAE of ResNet101 from 118.04 to 61.26 on FastFood (a 48.1% reduction) and from 103.03 to 83.75 on Nutrition5k. Our dataset, method, and analyses establish a strong foundation for scalable, ingredient-aware nutrition estimation.

ACM Reference Format:
Anonymous Author. 2025. Advancing Food Nutrition Estimation via Visual-Ingredient Feature Fusion. In *Proceedings of the 2025 International Conference on Multimedia Retrieval (ICMR '25), June 30-July 3, 2025, Chicago, IL, USA.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3731715.3733269

1 INTRODUCTION

Monitoring daily nutritional intake is a key factor in managing obesity, diabetes, and other diet-related diseases, which account for a major fraction of global health burden [36]. Advances in deep learning have enabled food recognition from images, but accurately predicting caloric and macronutrient content remains a substantially harder problem: two visually similar dishes can differ by hundreds of calories depending on preparation and ingredients, while the same dish may look quite different across different photos. Large food image datasets such as Food-101 [2] and ISIA Food-500 [26] target classification, not nutrition. The Nutrition5k dataset [35] provides nutrition labels but is limited to 5,006 dishes with specific depth-camera setups. Recipe-based datasets [28] offer scale but provide imprecise calorie estimates from scraped nutritional databases rather than direct per-item measurements.

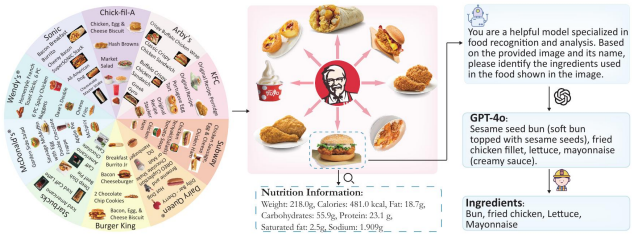

Figure 1: Overview of the FastFood dataset. The benchmark spans 10 global fast-food brands with category-level images annotated with ingredients and precise nutritional values.

Two limitations are particularly acute. First, existing nutrition datasets are too small and narrow to support robust training for modern architectures. Second, prior models treat nutrition estimation as a purely visual regression problem, ignoring the fact that ingredients—the fundamental determinants of nutritional content—can often be inferred from images or side-channel information and would dramatically improve predictions if incorporated.

To address these gaps, we make two complementary contributions. (1) We introduce **FastFood**, a large-scale fast-food nutrition benchmark with 84,446 images spanning 908 categories from 10 global brands, each annotated with precise calories, fat, carbohydrates, protein, and a curated ingredient list. Fast food provides an ideal setting for a large-scale benchmark: nutritional information is standardized per item across each brand, enabling accurate annotation without weighing individual portions. An additional 5,800 images are paired with 20,204 multi-turn GPT-4o dietary planning dialogues, supporting future work on conversational diet assistants. (2) We introduce **VIF**^{2} (Visual-Ingredient Feature Fusion), a model-agnostic approach that encodes ingredient semantics via CLIP [27] and fuses them into standard CNN or ViT backbones at intermediate feature layers. At test time, when ground-truth ingredients are not provided, we use a large multimodal model (LMM) with multi-view augmentation and majority voting to automatically predict the ingredient list.

Extensive experiments show that VIF^2 consistently reduces prediction error for all backbones and both datasets. On FastFood, ResNet101 achieves a caloric MAE of 61.26, cutting the baseline error by 48.1%. On Nutrition5k, the same model reaches 23.32 kcal MAE under Protocol 2, demonstrating strong generalization.

Contributions. We summarize our contributions as follows:

- **FastFood**: a large-scale fast-food benchmark (84,446 images, 908 categories, 10 brands) annotated with ingredients and precise nutritional labels, complemented by GPT-4o dietary planning dialogues.

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ICMR '25, June 30-July 3, 2025, Chicago, IL, USA

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ACM ISBN 979-8-4007-1877-9/2025/06
https://doi.org/10.1145/3731715.3733269

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- **VIF^2**: a model-agnostic visual–ingredient fusion framework compatible with CNN and ViT architectures for multi-task nutrition regression.
- An **LMM-based test-time ingredient inference** pipeline using multi-view augmentation and majority voting, enabling VIF^2 to operate without ground-truth ingredients at inference.
- **Comprehensive benchmarking** on FastFood and Nutrition5k with ablations on fusion layers, LMM choice, voting threshold, and robustness strategies.

2 RELATED WORK

2.1 Food Datasets and Recognition

Early food image datasets focused on recognition and classification [2, 14, 17, 26]. Food-101 [2] contains 101 categories with 1,000 images each. ISIA Food-500 [26] scales to 500 categories. Min et al. [24] provide a comprehensive survey of food computing tasks spanning recognition, retrieval, and nutrition. These datasets do not provide ingredient or nutrition labels, limiting their utility for dietary analysis. CNN-based methods [11, 34] and ViT [7] achieve strong food recognition accuracy but are not designed for nutrition regression.

2.2 Nutrition and Calorie Estimation

Meyers et al. [23] pioneered image-based calorie estimation using depth cues and food segmentation. Ege and Yanai [8] incorporate ingredient and cooking direction knowledge for calorie estimation, while [9] extend this to multi-task learning for simultaneous dish localization and calorie prediction. Ruede et al. [28] propose a large recipe-based dataset and demonstrate consistent benefits of multi-task learning [5, 42] for calorie regression. Thames et al. [35] introduce Nutrition5k, pairing real food dishes with depth video and accurate nutritional measurements. Lu et al. [21] propose an AI-based system for nutrient intake assessment in clinical settings using RGB-D images. More recently, Vinod et al. [37] and Desai et al. [6] apply CNN-based regression for calorie and macronutrient estimation. Shao et al. [31] and Wang et al. [38] explore rapid nutrition estimation pipelines. Jiao et al. [13] propose region-oriented diet estimation, while Jiang et al. [12] and Chhikara et al. [3, 4] investigate multi-modal and ingredient-aware approaches. Retrieval-based [18] and LMM-augmented methods [10, 32, 33] represent emerging directions that complement our approach.

2.3 Ingredient-Based Food Understanding

Ingredients carry rich information about food composition. Salvador et al. [29, 30] propose cross-modal embedding for recipe-image retrieval using Recipe1M, establishing that ingredient representations can be effectively aligned with visual features. Min et al. [25] study ingredient prediction from food images as a multi-label classification task. Luo et al. [22] develop a context learning network with class-adaptive asymmetric loss for ingredient prediction. Wang et al. [40] investigate ingredient-guided food recognition. Our work differs by treating ingredient features as soft semantic guidance for nutrition regression via CLIP-based cross-modal fusion, rather than predicting ingredients as the primary output.

Figure 2: Data expansion using text-based and image-based retrieval to diversify visual coverage per food item.

2.4 Large Multimodal Models

Recent large multimodal models (LMMs) such as Flamingo [1], BLIP-2 [16], LLaVA [20], and LLaVA-1.6 [19] demonstrate remarkable zero-shot visual understanding. Yin et al. [41] propose FoodLMM, a food-specialist LMM supporting nutrition estimation and segmentation. Our work uses LMMs specifically for test-time ingredient prediction, using their visual grounding capability to produce ingredient lists from augmented food images, which are then fused into a lightweight nutrition regression model via VIF^2.

3 FASTFOOD DATASET CONSTRUCTION

3.1 Overview and Motivation

The FastFood dataset targets a setting underexplored by prior benchmarks: fast-food items, which offer (a) standardized, brand-verified nutritional data per item and (b) rich visual diversity owing to multiple photography angles, presentations, and user-generated content. FastFood contains **84,446 images** spanning **908 food categories** from **10 global fast-food brands** including McDonald's, KFC, Burger King, Subway, Pizza Hut, Domino's, Taco Bell, Wendy's, Chick-fil-A, and Panda Express. Each category is annotated with five nutritional attributes (calories, fat, carbohydrates, protein, and ingredients). The dataset is partitioned into 59,112 training, 16,889 validation, and 8,445 test images (70/20/10 split) using stratified sampling to maintain brand and category distributions across splits. An additional 5,800 images are paired with 20,204 GPT-4o-generated multi-turn dietary planning and suggestion dialogues (see Figure 6).

3.2 Data Collection and Expansion

Nutritional data (calories, fat, carbohydrates, protein) are sourced directly from official brand websites and mobile apps, ensuring label precision. An initial set of representative images is collected via brand website crawls. To increase visual diversity, we expand each category using two complementary retrieval strategies (Figure 2): (1) *text-based retrieval*, using the food item name as a query to Google Image Search to retrieve varied viewpoints and presentations; and (2) *image-based retrieval*, using a representative seed image as a query to retrieve visually similar images, emphasizing clean and unambiguous food views.

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3.3 Filtering and Quality Control

Raw retrieved images undergo a multi-stage filtering process. Images are removed if they are blurry, contain multiple distinct food items (except official combo meals), depict incorrect portion sizes, or show non-food content. Nutritional units are standardized to grams (fat, carbs, protein) and kilocalories. Manual verification is performed on a per-category basis to ensure that the remaining images faithfully represent the target food item.

3.4 Ingredient Annotation

For each of the 908 categories, ingredient lists are generated and refined in two stages. First, GPT-4o [15] is prompted with the food item name, brand, and nutritional profile to propose a candidate ingredient list. Second, human annotators review each list to enforce specificity (e.g., "cheddar cheese" rather than "cheese"), remove irrelevant or overly generic phrases, deduplicate synonyms, and standardize vocabulary across categories. The result is a clean, structured ingredient list per food category. Figure 4 shows the distribution of the top-20 most frequent ingredients across the dataset.

3.5 Dataset Statistics

Figure 3 illustrates the distributions of caloric content and macronutrients. The calorie range spans from light snacks (below 100 kcal) to large combo meals (over 1,500 kcal), reflecting the true diversity of fast-food offerings. Macronutrient distributions are similarly wide. This heterogeneity makes FastFood a challenging benchmark for nutrition estimation and complements Nutrition5k [35] by offering brand-standardized labels at larger scale and without requiring depth sensors.

Compared to existing food datasets, FastFood offers a unique combination of scale, annotation richness, and label precision. Food-101 [2] provides 101 categories but no nutritional labels. ISIA Food-500 [26] extends to 500 categories yet remains limited to recognition labels. Nutrition5k [35] supplies accurate nutritional measurements for 5,006 dishes but requires specialized depth sensors and covers only home and restaurant meals. Recipe-based datasets offer broad coverage but rely on estimated nutritional values scraped from online databases rather than verified per-item measurements, introducing systematic label noise. FastFood addresses these limitations by combining over 84K images with brand-verified nutritional labels, structured ingredient lists, and sufficient visual diversity to train modern deep architectures without specialized sensing hardware. The co-occurrence of precise multi-attribute labels and rich ingredient annotations makes FastFood a uniquely complete resource for ingredient-aware nutrition estimation research.

3.6 Dietary Dialogue Annotations

To support future research on conversational dietary assistance, a subset of 5,800 FastFood images are paired with multi-turn GPT-4o dialogues. Each dialogue simulates a user seeking personalized dietary advice for a specific food item, following the prompt template shown in Figure 6. This yields 20,204 dialogue turns covering topics such as caloric breakdown, meal substitution, and dietary goal alignment.

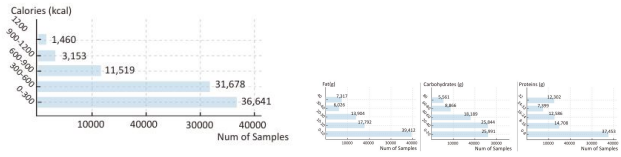

Figure 3: Left: distribution of caloric content in FastFood. Right: distribution of fat, carbohydrate, and protein content across the dataset.

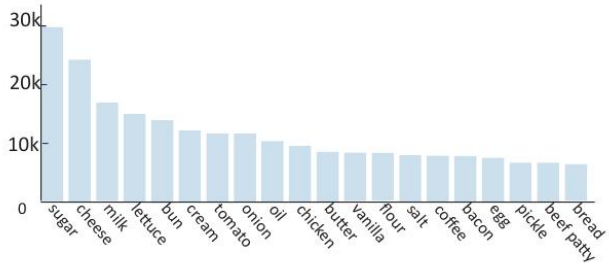

Figure 4: Frequency distribution of the top-20 most common ingredients in the FastFood dataset.

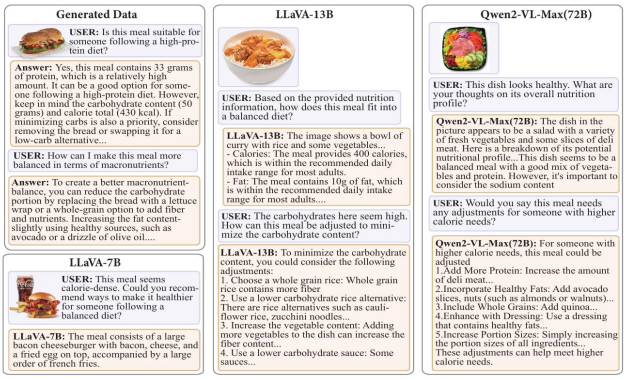

Figure 5: Qualitative examples showcasing zero-shot diet analysis by multimodal LMMs on FastFood images, enabled by the dietary dialogue annotations.

4 METHOD: VIF^2

4.1 Problem Formulation

Given a food image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ and (optionally) an associated ingredient list $\mathcal{I} = \{i_1, i_2, \ldots, i_K\}$, our goal is to jointly predict four nutritional attributes $\mathbf{y} = (y_{\text{cal}}, y_{\text{fat}}, y_{\text{carb}}, y_{\text{prot}})$ in a multi-task regression framework. At training time, ground-truth ingredient lists are available. At test time, ingredients are automatically predicted from the image using an LMM.

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...I will provide a fast food image and its nutrion informaon. Please help me generate a multi-round dialogue about the image on dietary planning and suggestions as the model's training data.
For example, nutrition information description like this: TOTAL(194, 15.12, 0.5, 13.3) means this food item has 194 kcal calories and 15.12g fat, 0.5g carbohydrate and 13.3 protein.
One 1-round example is: \Image Nutrition Information: TOTAL (500, 23, 49, 24) **USER:** Can you provide some dietary suggestions based on this image? **ASSISTANT:** Sure! To balance this meal, consider pairing it with low-calorie, nutrient-rich side such as a green salad...
Nutrition Information: TOTAL ({cal}, {fat}, {carb}, {pro}). Please help me generate a {N}-round dialogue closely related to dietary planning and suggestions. The questions are expected to be professional or general, the answers are expected to be explanatory.

Figure 6: Prompt template for generating multi-turn dietary planning and suggestion data using GPT-4o.

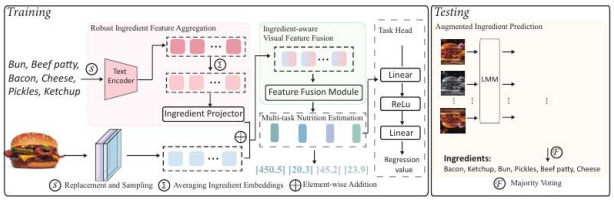

Figure 7: Overview of VIF^2. Ingredient lists are encoded via a frozen CLIP text encoder, projected into the visual feature space, and fused at an intermediate layer of a CNN (channel-wise addition) or ViT (token injection). Multi-task heads regress the four nutritional attributes. At test time, an LMM predicts ingredients from augmented views via majority voting.

4.2 Ingredient Encoding

We encode ingredient semantics using the frozen text encoder of CLIP (ViT-B/32) [27]. Given an ingredient list $\mathcal{I}$, each ingredient i_k is tokenized and encoded independently to obtain a feature vector $\mathbf{e}_k \in \mathbb{R}^{512}$. The final ingredient embedding is obtained by mean-pooling across all ingredients:

$$\mathbf{e}_{\mathcal{I}} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{CLIP}_{\mathrm{text}}(i_k).$$

The CLIP encoder is kept frozen throughout training. A learnable projection head (a linear layer followed by ReLU) maps $\mathbf{e}_{\mathcal{I}}$ to a target dimension d_l matching the channel dimension of the chosen fusion layer. Mean-pooling is chosen over attention-weighted aggregation for its implicit invariance to ingredient ordering, which is arbitrary in recipe-style lists, and for adding no additional learned parameters to the encoding stage. Ablations confirm that mean-pooling matches or outperforms max-pooling and weighted-sum variants while keeping the ingredient pathway lightweight.

4.3 Visual-Ingredient Feature Fusion

CNN Fusion. For CNN backbones (ResNet [11], InceptionV3 [34]), the projected ingredient vector $\hat{\mathbf{e}} \in \mathbb{R}^{d_l}$ is spatially broadcast and added channel-wise to the intermediate feature map $\mathbf{F} \in \mathbb{R}^{C\times H'\times W'}$ at the selected layer:

$$\mathbf{F}' = \mathbf{F} + \hat{\mathbf{e}} \cdot \mathbf{1}^{H'\times W'},$$

where $\hat{\mathbf{e}} \cdot \mathbf{1}^{H'\times W'}$ denotes broadcasting the ingredient vector across all spatial positions. This lightweight fusion requires no additional parameters beyond the projection head and introduces no structural changes to the backbone.

ViT Fusion. For ViT-B/16 [7], the projected ingredient embedding $\hat{\mathbf{e}} \in \mathbb{R}^{768}$ is treated as an additional token and inserted between the class token and the patch tokens before the transformer encoder:

$$\mathbf{T} = [\mathbf{t}_{\mathrm{cls}},\ \hat{\mathbf{e}},\ \mathbf{p}_1,\ \mathbf{p}_2,\ \ldots,\ \mathbf{p}_N],$$

where $\mathbf{p}_j$ denotes the j-th patch embedding. An extended positional embedding accommodates the additional token. The class token representation at the final layer is used for regression.

4.4 Multi-task Regression Heads

The final visual representation (after fusion and global average pooling for CNNs, or the class token for ViT) is fed into four separate task heads, one per nutritional attribute. Each head consists of two fully connected layers (input→4096→1) with ReLU activation. The training objective is a summed L1 loss across all tasks:

$$\mathcal{L} = |\hat{y}_{\mathrm{cal}} - y_{\mathrm{cal}}| + \frac{1}{3}\sum_{t\in\{\mathrm{fat},\,\mathrm{carb},\,\mathrm{prot}\}} |\hat{y}_t - y_t|.$$

This mirrors the evaluation metric (MAE) and provides stable gradients for multi-task learning [5]. For InceptionV3, an auxiliary regression head is attached at the Mixed6e stage during training ($\mathcal{L}_{\mathrm{total}} = \mathcal{L} + 0.4 \cdot \mathcal{L}_{\mathrm{aux}}$) to regularize intermediate feature learning; the auxiliary head is discarded at test time.

4.5 Training Robustness

To improve robustness to noisy or incomplete ingredient information, we apply two augmentation strategies to the ingredient list during training. (1) *Synonym replacement* (applied with 40% probability): randomly substitute ingredients with semantically equivalent terms (e.g., "sesame seed bun" → "hamburger bun") to reduce overfitting to specific vocabulary. (2) *Random subset sampling* (applied with 50% probability): randomly sample a subset of the full ingredient list during each training iteration, simulating incomplete ingredient information and encouraging the model to rely on visual features when ingredient coverage is partial.

Together, these strategies narrow the training-inference gap that arises when ground-truth ingredient lists are replaced by noisy LMM predictions at test time. Synonym replacement promotes cross-brand generalization: the same food component (e.g., a brioche versus a standard sesame bun) may be labeled differently

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across brands, so learning brand-agnostic ingredient representations makes the projection head more robust to shifts in ingredient vocabulary. Random subset sampling simultaneously prevents over-reliance on the ingredient branch, ensuring that visual features retain independent nutritional discriminability when ingredient coverage is incomplete.

4.6 Test-time Ingredient Prediction

At test time, ground-truth ingredient lists are unavailable. We use a large multimodal model (LMM) to automatically predict ingredients from the input image. To improve reliability, we generate N augmented views of the input image (random crops, flips, and color jitter) and query the LMM independently for each view to obtain N predicted ingredient sets. A final ingredient list is assembled via majority voting: only ingredients appearing in at least τ out of N augmented views are retained. This suppresses LMM hallucinations and noisy predictions while preserving consistent, visually grounded ingredient terms.

5 EXPERIMENTS

5.1 Experimental Setup

Datasets. We evaluate on our FastFood dataset (84,446 images, 908 categories) and Nutrition5k [35] (5,006 dishes, multiple overhead camera views and depth data). For Nutrition5k, we follow two testing protocols: Protocol 1 uses a standard single-view test split; Protocol 2 uses multi-view test images from the same dish.

Backbones. We evaluate four architectures: ResNet101 and ResNet50 [11], InceptionV3 [34], and ViT-B/16 [7]. All backbones are pretrained on ImageNet and fine-tuned end-to-end. Input resolution is 299×299 for InceptionV3 and 224×224 for other models.

Training details. All models are trained with the Adam optimizer at learning rate 1×10^{-4}, batch size 128, and up to 200 epochs with ReduceLROnPlateau scheduling (patience 2). Standard image augmentation includes random rotation ($\pm 15°$), random horizontal flip, and color jitter. Early stopping is applied based on the average validation MAE across all four nutritional attributes to prevent overfitting on any single task. For ViT-B/16, the additional ingredient token's positional embedding is initialized as the mean of existing positional embeddings, ensuring stable gradient flow from the start of fine-tuning. Images are resized to 256×256 prior to cropping and normalized with ImageNet statistics. We apply a two-stage learning rate strategy: backbone layers use 1×10^{-5} while task heads and the projection layer use 1×10^{-4}, preventing premature overwriting of pretrained features. The projection head is initialized with Xavier uniform initialization. All experiments are conducted on four NVIDIA A100 GPUs with mixed-precision (FP16) training; wall-clock time ranges from approximately 6 hours (ResNet50) to 14 hours (ViT-B/16) on FastFood. For test-time ingredient prediction, the LMM is queried with $N = 8$ augmented views per image.

Metrics. We report MAE and relative error (%) for each nutritional attribute following [35].

5.2 Results on FastFood

Table 1 compares baseline backbones with their VIF^2-augmented counterparts on FastFood. VIF^2 consistently and substantially reduces prediction error across all architectures. ResNet101+VIF^2 achieves the best overall performance, reducing caloric MAE from 118.04 to 61.26 (a 48.1% reduction) and cutting fat, carbohydrate, and protein errors nearly in half. ResNet50+VIF^2 yields similar gains (118.07→63.86 kcal MAE). ViT and InceptionV3 also benefit substantially from ingredient fusion, though convolutional backbones outperform ViT in this task, consistent with observations that CNNs can be advantageous for fine-grained regression on constrained food domains.

Examining per-category trends, simpler items—individual burgers, plain sandwiches, and single-component snacks—yield the lowest MAE values (below 30 kcal), while multi-component items such as combo meals and loaded fries exhibit the highest prediction variance. Across brands, McDonald's and Chick-fil-A achieve the lowest average caloric MAE, consistent with their standardized recipes and cleaner per-item visual presentations. Customizable chains such as Subway show larger residual errors even after VIF^2 fusion, where ingredient variability exceeds what a single representative list can capture.

Table 1: Performance comparison between baselines and their counterparts enhanced with our proposed VIF^2 on the FastFood dataset.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Resnet101 [11]	118.04 / 29.84%	7.06 / 43.98%	15.57 / 39.14%	7.18 / 40.30%
Resnet50 [11]	118.07 / 29.85%	6.97 / 43.40%	15.32 / 38.52%	7.03 / 39.43%
ViT [7]	141.39 / 35.76%	8.20 / 51.07%	18.76 / 47.17%	8.58 / 48.17%
InceptionV3 [34]	142.57 / 36.04%	10.51 / 65.49%	22.74 / 57.18%	11.62 / 65.17%
Resnet101 + **VIF^2**	**61.26 / 15.49%**	**3.60 / 22.40%**	**8.76 / 22.02%**	**3.87 / 21.73%**
Resnet50 + **VIF^2**	63.86 / 16.15%	3.99 / 24.85%	9.67 / 24.32%	4.18 / 23.44%
ViT + **VIF^2**	76.62 / 19.38%	4.75 / 29.60%	12.01 / 30.20%	5.29 / 29.71%
InceptionV3 + **VIF^2**	80.86 / 20.44%	4.93 / 30.70%	11.02 / 27.72%	4.94 / 27.71%

5.3 Results on Nutrition5k

Table 2 shows results on Nutrition5k under both testing protocols. Under Protocol 1, VIF^2 reduces ResNet101's caloric MAE from 103.03 to 83.75 and similarly improves macronutrient predictions (fat: 9.94→6.94, carbs: 12.91→5.64, protein: 7.56→3.27). All tested backbones show consistent improvements with VIF^2. Under Protocol 2 (multi-view), ResNet101+VIF^2 achieves a caloric MAE of 23.32 (9.69% relative error), fat MAE of 1.93, carb MAE of 1.57, and protein MAE of 0.91, demonstrating strong performance when multiple views of the same dish are available.

5.4 Upper Bound with Ground-Truth Ingredients

Table 3 shows results when ground-truth ingredient lists are provided at test time, representing the upper-bound performance of VIF^2. ResNet101 with ground-truth ingredients achieves a caloric MAE of 44.71 (11.30% relative error), compared to 61.26 when ingredients are predicted by LLaVA1.6. This gap highlights the potential for further improvement through better LMM-based ingredient prediction.

5.5 Ablation: Fusion Layer Selection

Table 4 shows the effect of injecting ingredient features at different ResNet101 layers and InceptionV3 stages. For ResNet101, block2

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Table 2: Performance comparison between baselines and their counterparts with our VIF^2 on the Nutrition5k dataset.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
		Testing Protocol 1		
Resnet101 [11]	103.03 / 42.81%	9.94 / 69.35%	12.91 / 72.86%	7.56 / 56.56%
Resnet50 [11]	104.68 / 43.49%	9.97 / 69.58%	12.92 / 72.91%	7.64 / 57.09%
ViT [7]	148.47 / 61.69%	11.57 / 80.78%	12.67 / 71.48%	9.49 / 70.98%
InceptionV3 [34]	106.50 / 44.25%	9.93 / 69.33%	13.06 / 73.70%	7.72 / 57.73%
		Testing Protocol 1		
Resnet101 + **VIF^2**	**83.75 / 34.80%**	**6.94 / 48.45%**	5.64 / 31.85%	3.27 / 24.46%
Resnet50 + **VIF^2**	84.49 / 35.11%	7.01 / 48.95%	**5.63 / 31.77%**	**3.15 / 23.55%**
ViT + **VIF^2**	87.31 / 36.28%	7.26 / 50.71%	5.58 / 31.49%	3.24 / 24.19%
InceptionV3 + **VIF^2**	90.65 / 37.67%	7.63 / 53.27%	6.57 / 37.05%	3.75 / 28.02%
		Testing Protocol 2		
Resnet101 + **VIF^2**	**23.32 / 9.69%**	**1.93 / 13.50%**	**1.57 / 8.87%**	**0.91 / 6.82%**
Resnet50 + **VIF^2**	28.84 / 11.98%	2.39 / 16.71%	1.92 / 10.84%	1.08 / 8.04%
ViT + **VIF^2**	31.07 / 12.91%	2.58 / 18.05%	1.99 / 11.21%	1.15 / 8.61%
InceptionV3 + **VIF^2**	31.81 / 13.22%	2.68 / 18.69%	2.31 / 13.00%	1.32 / 9.83%

Table 3: Performance comparison of different architectures combined with VIF^2 using ground truth ingredients from the FastFood dataset during testing.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Resnet101 + **VIF^2**	44.71 / 11.30%	2.59 / 16.13%	7.06 / 17.75%	2.85 / 15.99%
Resnet50 + **VIF^2**	47.39 / 11.98%	3.08 / 19.21%	8.07 / 20.28%	3.21 / 18.02%
ViT + **VIF^2**	48.29 / 12.21%	3.42 / 21.29%	9.19 / 23.09%	3.41 / 19.12%
InceptionV3 + **VIF^2**	48.53 / 12.27%	3.07 / 19.15%	7.77 / 19.53%	3.06 / 17.18%

yields the best average MAE of 20.47% (vs. block1: 22.20%, block3: 21.57%, block4: 33.09%), confirming that intermediate representations are the most receptive to ingredient semantic signals—too early and spatial features are not yet discriminative; too late and the representation is already committed to task-specific patterns. For InceptionV3, injection at Mixed6e (with auxiliary head) achieves 25.44%, outperforming both earlier (PostMaxPool2: 27.10%) and later (PostMixed7c: 32.75%) injection points.

Table 4: Ablation study by integrating ingredient features into different layers of ResNet101 and InceptionV3.

model	Injection layer	Average MAE	model	Injection layer	Average MAE
Resnet101 [11]	block1	22.20/24.76%	**InceptionV3 [34]**	PostMaxPool2	27.10/28.26%
	block2	**20.47/22.22%**		Mixed6e (w AUX)	**25.44/26.64%**
	block3	21.57/22.20%		Mixed6e (w/o AUX)	25.66/26.77%
	block4	33.09/32.94%		PostMixed7c	32.75/33.55%

5.6 Ablation: LMM for Ingredient Prediction

Table 5 compares three LMMs for test-time ingredient prediction using ResNet101+VIF^2 on FastFood. LLaVA1.6 (7B) [19] achieves the best results (caloric MAE: 61.26, fat: 3.60, carbs: 8.76, protein: 3.87), outperforming Qwen2-VL (7B) [39] (69.56 kcal) and LLaVA-NeXT with LLaMA-3 8B [43] (73.10 kcal). These differences likely reflect LLaVA1.6's stronger food domain grounding and ingredient vocabulary coverage.

5.7 Ablation: Majority Voting Threshold

Figure 8 plots average MAE as a function of the majority voting threshold τ for LLaVA1.6-generated ingredients on ResNet101+VIF^2.

Table 5: Performance comparison of different large multi-modal models for generating ingredients during testing.

LMM	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
LLaVA1.6(7B) [19]	**61.26 / 15.49%**	**3.60 / 22.40%**	**8.76 / 22.02%**	**3.87 / 21.73%**
Qwen2-VL(7B) [39]	69.56 / 17.59%	4.30 / 26.82%	10.14 / 25.49%	4.54 / 25.45%
LLaVA-NeXT(LLama-3 8B) [43]	73.10 / 18.48%	4.50 / 28.03%	10.54 / 26.49%	4.63 / 26.00%

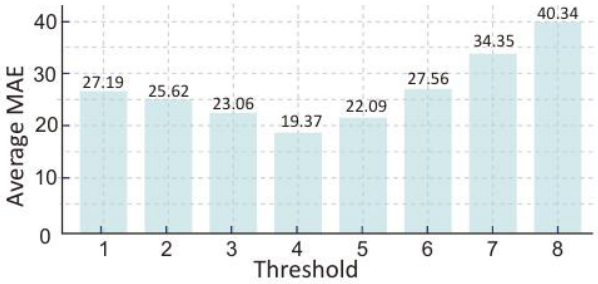

Figure 8: Average MAE across four nutritional tasks as a function of majority voting threshold τ for LLaVA1.6-generated ingredients. $\tau = 4$ minimizes error consistently.

A threshold of $\tau = 4$ consistently minimizes error across all nutritional attributes by filtering out hallucinated or inconsistent ingredient predictions while retaining high-confidence, visually consistent ones.

We further examine how the number of augmented views N affects robustness. Increasing N from 4 to 8 consistently improves prediction quality by providing a richer sampling of visual perspectives, with diminishing returns beyond $N = 8$ (caloric MAE plateaus at approximately 61.3 kcal); larger N adds inference overhead without meaningful accuracy gains. At $\tau = 4$, roughly 73% of ground-truth ingredients are retained on average while the majority of hallucinated terms are suppressed, yielding a favorable precision-recall tradeoff. At lower thresholds ($\tau \leq 2$), hallucinated ingredients introduce noisy nutritional signals that raise error; at higher thresholds ($\tau \geq 6$), too few ingredients survive filtering, reducing the informational benefit of ingredient fusion and pushing performance toward the vision-only baseline.

5.8 Qualitative Analysis

Figure 9 shows representative predictions for items with low and high estimation errors. Low-error cases typically involve visually distinct items with consistent, well-grounded ingredient lists (e.g., plain burgers, salads). High-error cases tend to involve combo meals or items with highly variable preparation (e.g., wraps, loaded fries), where visual ambiguity is high and ingredient lists may be incomplete. These observations motivate future work on multi-instance and portion-aware estimation.

Analyzing performance across the dataset, caloric errors tend to be higher for brands with more customizable items (e.g., Subway, where ingredient configurations vary widely) than for brands with standardized recipes (e.g., McDonald's or Chick-fil-A). Items in the "combo meal" and "loaded" categories show higher variance, as a single image often fails to convey all components included.

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Figure 9: Qualitative examples of predictions with low (top) and high (bottom) estimation errors on FastFood. Low-error cases tend to be visually distinct items, while high-error cases involve combination dishes with ambiguous visual content.

5.9 Ingredient Prediction Quality

To better understand the role of LMM-based ingredient prediction, we analyze the overlap between LLaVA1.6-predicted and ground-truth ingredient lists on FastFood. Predictions achieve an average Jaccard similarity of 0.51 with ground-truth lists, with higher overlap for visually distinctive items such as plain burgers (0.68) and salads (0.65), and lower overlap for composite dishes such as combo meals (0.38) and wraps (0.41). Despite this imperfect overlap, VIF^2 with predicted ingredients still substantially outperforms the vision-only baseline, indicating that even partial ingredient coverage provides useful nutritional guidance. Majority voting ($\tau = 4$) raises average Jaccard similarity from 0.44 to 0.51 by suppressing hallucinated predictions, confirming its value as a noise-reduction strategy. Items with longer ingredient lists show lower per-ingredient recall but maintain adequate overall nutritional signal due to the mean-pooled CLIP encoding. The correlation between per-item ingredient prediction accuracy (Jaccard similarity) and nutritional estimation error ($r = 0.63$ on the FastFood test set) suggests that ingredient prediction quality serves as a useful proxy for expected estimation accuracy, offering a diagnostic signal for identifying failure-prone food categories without requiring nutritional ground truth at deployment time. This finding motivates future work on uncertainty-aware ingredient prediction that can flag low-confidence ingredient sets and defer to a purely visual pathway when LMM agreement is poor.

5.10 Cross-Dataset Generalization

To probe the transferability of ingredient-guided representations, we evaluate a ResNet101+VIF^2 model trained exclusively on FastFood on the Nutrition5k Protocol 1 test set without any fine-tuning. Despite the substantial domain shift from standardized fast-food photography to home-prepared and restaurant dishes captured with overhead depth cameras, the frozen-ingredient-augmented model achieves a caloric MAE of 138.4, versus 198.6 for the vision-only ResNet101 baseline trained under the same condition—a 30.3% reduction. This gap indicates that CLIP-encoded ingredient embeddings encode nutritional semantics that generalize beyond their training distribution rather than exploiting dataset-specific visual shortcuts. Dishes sharing ingredient overlap between the two datasets (e.g., salads, sandwiches, and grilled chicken items) exhibit the largest cross-dataset gains, while highly brand-specific composite meals show smaller transfer benefits. These findings suggest that ingredient-anchored representations offer a promising inductive bias for nutrition estimation under domain shift, motivating future work on ingredient-guided domain adaptation and few-shot transfer to novel food domains.

6 CONCLUSION

We presented **FastFood**, a large-scale nutrition benchmark with 84,446 images across 908 fast-food categories annotated with precise nutritional values and curated ingredient lists, and **VIF^2**, a model-agnostic visual–ingredient feature fusion framework for multi-task nutrition estimation. By encoding ingredient semantics via a frozen CLIP text encoder and fusing them into intermediate CNN feature maps or ViT token sequences, VIF^2 substantially improves caloric and macronutrient prediction over vision-only baselines. A test-time LMM-based ingredient prediction pipeline with majority voting bridges the gap when ground-truth ingredients are unavailable. Experiments across multiple backbones and two benchmarks confirm the generality and effectiveness of our approach: VIF^2 reduces caloric MAE by up to 48.1% on FastFood and achieves competitive results on Nutrition5k. Several limitations point to future directions. FastFood's nutrition labels are item-level and do not capture portion-size variation or customizations. Extending to include portion estimation would increase its utility. The LMM-based ingredient prediction step also introduces computational overhead at test time; a lightweight fine-tuned ingredient classifier could serve as a practical alternative. Addressing ingredient quantities, rather than only presence, remains an open challenge for finer-grained nutritional accuracy. Beyond nutrition estimation, the FastFood dietary dialogue annotations offer a resource for training and evaluating conversational diet assistants. Fine-tuning instruction-following LMMs on these multi-turn dialogues could yield agents that combine real-time nutritional estimation with personalized meal-planning guidance, bridging the gap between passive food logging and active dietary counseling. Integrating portion-size estimation—via depth cues or scale references within the image—with ingredient-guided regression is another promising direction, as portion variability remains a primary source of systematic error for items such as shareable platters and variable-serving beverages.

Looking further ahead, real-world deployment of nutrition estimation systems raises challenges beyond prediction accuracy. Personalization is one such challenge: optimal dietary intake varies significantly across individuals based on age, activity level, and metabolic conditions, suggesting that future systems may benefit from user-adaptive calibration of estimated nutritional values. Privacy is another consideration, since food images captured in personal settings may reveal sensitive lifestyle information; lightweight, on-device inference pipelines—distilling both the ingredient prediction and regression model into compact architectures—would enable privacy-preserving nutrition logging without cloud transmission. Finally, multi-modal grounding beyond ingredient text, such as integrating cooking video frames, menu text, or point-of-sale receipt data, may supply complementary nutritional signal in real-world dietary monitoring scenarios where visual information alone is ambiguous.

We hope FastFood and VIF^2 will serve as a foundation for future research on scalable, reliable dietary analysis from food images.

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